SIBONGAKONKE BUTHELEZI

SOFTWARE ENGINEER

JOHANNESBURG | 0664484226 | sibongakonkebuthelezi91@gmail.com | https://github.com/sibongakonkebuthelezi91

Professional Overview

I am a Software Developer with hands-on experience building web applications, backend services, database-driven systems, and containerized applications. My strongest foundation is in Python, with practical experience using Flask, Django, SQLite, JavaScript, Docker, Git, and CI/CD pipelines. I enjoy understanding how a system works from end to end, from gathering requirements and designing the user experience to developing backend logic, working with databases, testing, and deployment. My background in Business Management also allows me to look at software from a broader business perspective. I think about how technology fits into existing workflows, how users will interact with a system, and whether the solution is actually solving the problem it was designed to solve. I have experience developing role-based portals, business management applications, command-line tools, and a containerized messaging service. I am comfortable working independently as well as collaborating with other developers and stakeholders. I enjoy breaking down problems, debugging issues, learning unfamiliar technologies, and improving solutions through feedback. I am particularly interested in growing my skills in backend development, software architecture, DevOps, APIs, and cloud-based systems while contributing to software that creates real value.

Experience

Freelance Full-Stack Developer

CareerSine **04 July 2026 - present**

- I am currently developing a private business management web application for CareerSine, working directly with the client to understand their operational requirements and translate them into a practical software solution.
- I am responsible for the development process from requirements analysis and system planning through to implementation, database design, user interface development, testing, and ongoing improvements.
- The application is being developed using Django, JavaScript, and SQLite, with a focus on helping the business manage internal operations more efficiently.
- I work with Django's ORM to structure and manage application data while developing backend functionality and database relationships that support the client's workflows.
- I use JavaScript to create interactive functionality and improve the responsiveness and usability of the application.
- Because this is a client-facing project, I regularly communicate progress, clarify requirements, demonstrate completed functionality, and use feedback to refine the system.
- This experience has strengthened both my technical development skills and my ability to communicate technical decisions in a way that makes sense to non-technical stakeholders.

Projects

Campus & Student Management Portal

***Flask, Python, SQLite, HTML, CSS* | https://github.com/bhekani17/wethinkcode-branch-register**

- I developed a web-based campus and student management system designed to support administrative processes such as student registration, check-ins, and user management.
- I also worked with SQLite for persistent data storage and developed validation logic to handle information such as student IDs, email addresses, passwords, and other user inputs.
- I implemented authentication and session management for different types of users and developed role-based access controls to separate administrative functionality from student functionality.
- I designed custom validation modules to ensure data integrity for student IDs, emails, passwords, and other user inputs.

Binary Search Number Guessing Game

Python | https://github.com/sibongakonkebuthelezi91/practice-activity

- I developed an interactive command-line game in Python that uses the binary search algorithm to efficiently determine a number selected by the user.
- Instead of checking numbers sequentially, the program continually reduces the possible search range based on the user's feedback, giving the algorithm an O(log n) search complexity.
- I structured the program into separate functions for input validation, game logic, user interaction, and replay functionality.
- The project helped strengthen my understanding of algorithms, computational efficiency, modular programming, and writing code that is easier to maintain and extend.

Simple Mail Protocol

Flask, Python 3.12, Docker | https://github.com/sibongakonkebuthelezi91/simple_mail_protocol

- I developed a Flask-based web application that accepts email addresses through a web form, validates user input, and processes email requests through a structured backend workflow.
- I integrated the application with the Brevo email API using Python's requests library, allowing the system to send HTML-based email content to submitted email addresses.
- I separated application responsibilities across different Python modules, using dedicated functionality for web routing, email processing, and input validation to keep the application organised and maintainable.
- I implemented environment-based configuration using python-dotenv and environment variables so that API credentials and sensitive configuration are kept outside the application source code.

Education

Certificate in Business Management(N6) **2023**

My Business Management qualification gives me a useful perspective when developing software because I understand that technology needs to support real business processes and user requirements. It has helped me develop stronger communication, organisational, problem-solving, and stakeholder-management skills, which I apply when gathering requirements and translating business needs into practical technical solutions. I use this knowledge when thinking about how software can improve workflows, reduce unnecessary manual processes, and create better experiences for the people using a system.

Skills & abilities

- Languages: Python, Java, JavaScript, HTML5, CSS3, C#
- Frameworks & Libraries: Django, Flask, Django REST Framework, React (Basic)
- Databases: SQLite, PostgreSQL, Django ORM
- DevOps & Deployment: Docker, Docker Compose, CI/CD Pipelines, Environment Configuration
- Developer Tools: Git, GitHub, GitLab, VS Code, PyCharm, IntelliJ IDEA, Linux Command Line
- Core Competencies: RESTful APIs, User Authentication, Role-Based Access Control (RBAC), Data Validation, Object-Oriented Programming (OOP)
- Problem-Solving: Debugging, Troubleshooting, Testing
- Soft Skills: Agile Methodologies, Team Collaboration, Client Communication, Requirements Gathering